A Data Driven Analysis of 30 Years of Pokémon

What will the gen10 Pokédex size be? Is powercreep real? Are water types bulkier on average or is Suicune gaslighting us? To answer these types of questions I will apply data science concepts and methods to Pokémon game data, such as base stats, types, etc. For the machine learning part of the project, I have two ideas: predicting the generation 10 Pokédex size as well as the type distribution of the generation or attempting to create an Akinator like model to guess any Pokémon. I will collect my data from PokéAPI, an open-source project with 182 contributors. I will query this API through a python script, a simple proof-of-concept script can be found in my GitHub repository with the name: example_api_use.ipynb

Website: https://pokeapi.co/

PokéAPI GitHub Repo: https://github.com/PokeAPI/pokeapi/

There is a plethora game data available from PokéAPI, this project will use a subset of this data for example Pokédex entries are not useful for my project. I expect to have 16 columns for each data entry (subject to change). The fields can be found below:

- NationalDexID - int
- Name - string
- Type Primary - string
- Type Secondary - string
- Abilitiy 1 - string
- Abilitiy 2 - string
- Ability Hidden - string
- HP - int
- ATK - int
- DEF - int
- SATK - int
- SDEF - int
- SPD - int
- BST (Base stat total) - int
- Evolution Stage - int
- Generation – int